

DATA SCIENCE CASE STUDY

Titanic

Survival Analysis

Python | Pandas | NumPy | Matplotlib | Seaborn

 Reference Report: titanic_full_narrative_report.pdf

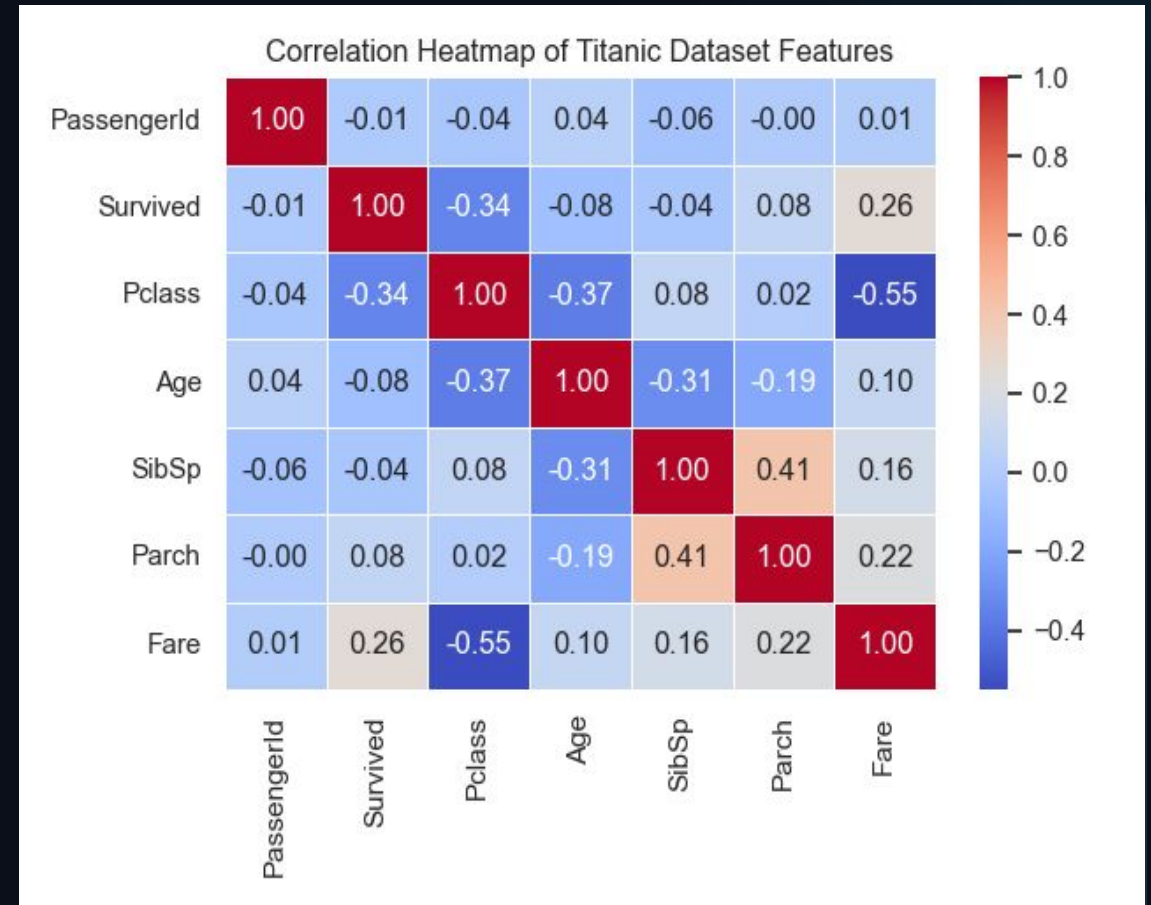
Key Questions

- Did gender affect survival?
- Did passenger class matter?
- How did age and fare influence outcomes?

Feature Correlations

- **Negative Correlation with Class:** Passenger class (Pclass) exhibits a strong negative correlation (-0.34) with survival rates.
- **Fare vs Survival:** Economic spending shows a positive relationship (+0.26), signaling a bias towards saving high-paying customers.
- **Age Factor:** Age demonstrates relatively low linear correlation (-0.08) with binary survival, suggesting non-linear survival structures.

Plot Name: correlation_heatmap.png

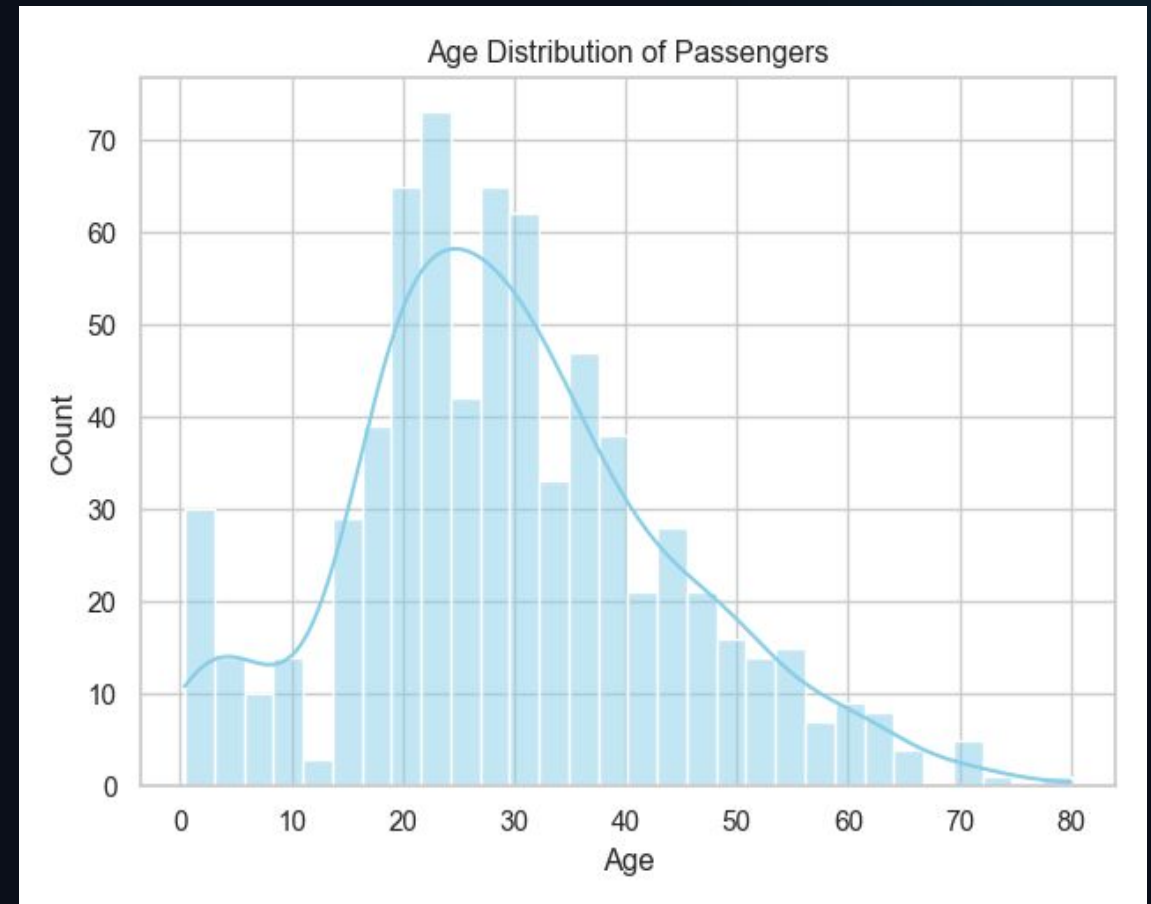


Passenger Age Distribution

- **Peak Demographics:** The majority of passengers on board were young adults, centered around 20 to 35 years old.
- **Child Survival Rates:** Children under 10 years old show a clear segment of survival support, aligning with the historical priority of "women and children first."
- **Elderly Mortality:** Advanced ages exhibit sparse data density with very low survival rates.

Chart: age_distribution.png

Boxplot: age_boxplot_by_survival.png

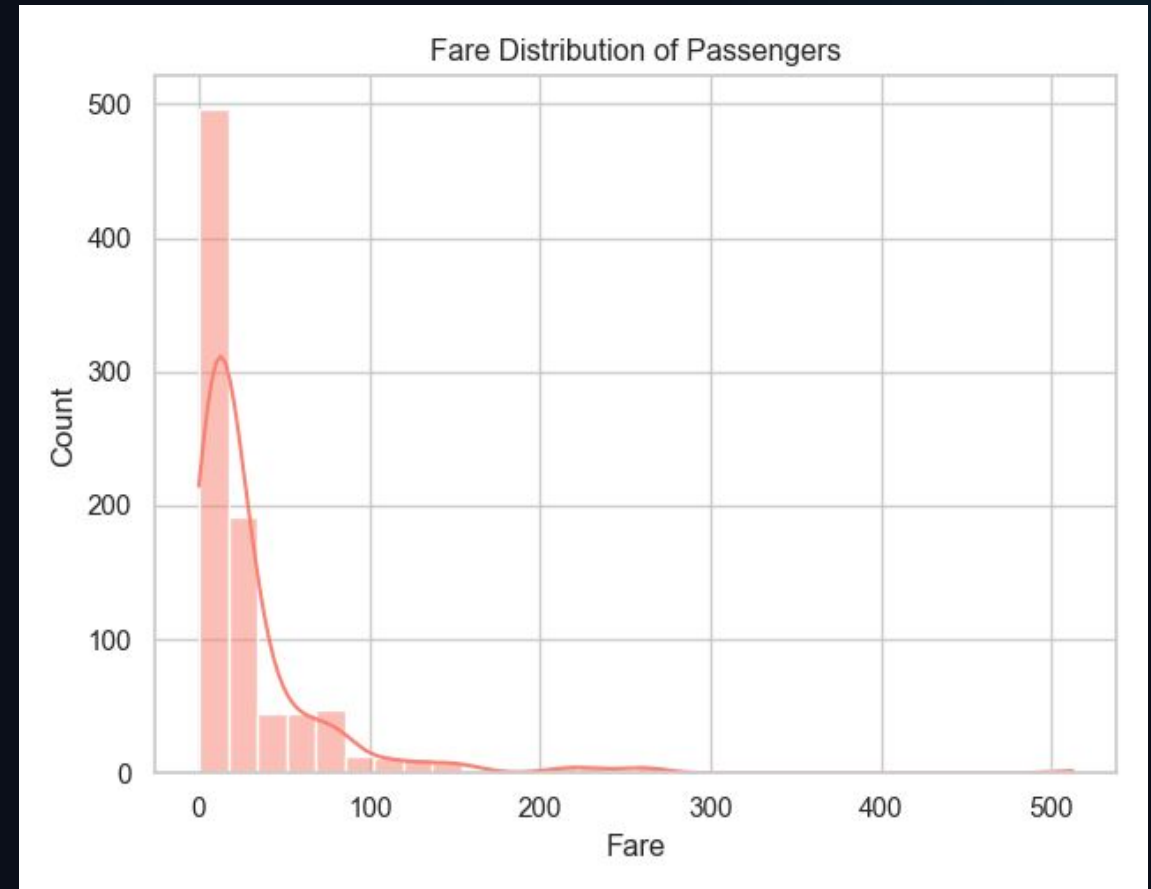


Fare Distribution Analysis

- **Economic Stratification:** Extreme right skew indicates that the vast majority paid nominal values (under £30), while a select few spent exorbitant sums.
- **Class Partitioning:** Class 1 fares cover an incredibly broad spectrum, pushing the scale limit up to £512.
- **Survival Alignment:** High-fare cohorts enjoyed significantly higher density in the survivor bracket.

📊 chart: fare_distribution.png

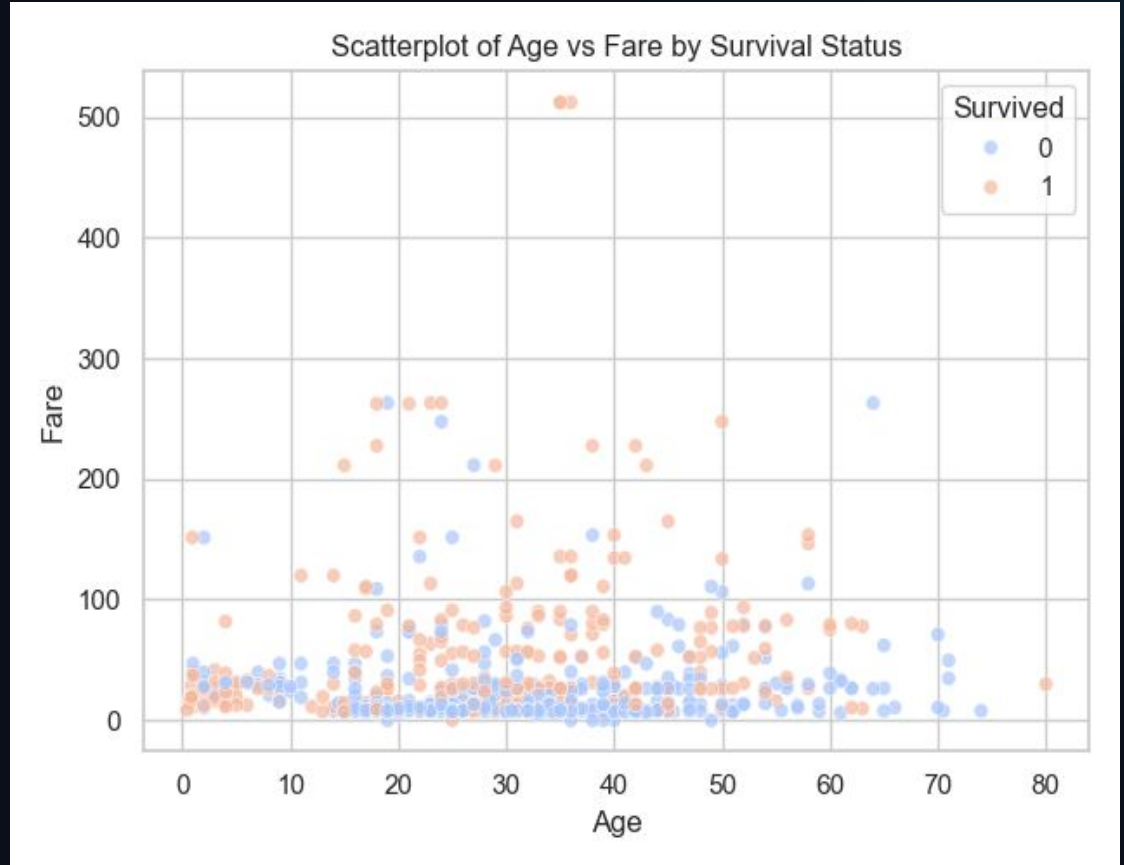
📊 boxplot: fare_boxplot_by_pclass.png



Age & Fare Relationship

- **Wealth Protection:** Nearly all passengers who paid premium fares (£200+) survived, regardless of age.
- **The Third-Class Squeeze:** Lower fares show high mortality concentrations, densely overlapping between ages 18 and 40.
- **Demographic Outliers:** Notable outliers include younger children in premium cabins who successfully navigated the crisis.

Scatterplot: age_fare_scatterplot.png



Key Survival Metrics



74.2

%

Female Passenger Survival
Rate



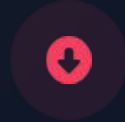
81.1%

Male Passenger Mortality
Rate



62.9%

First-Class Passenger
Survival Rate



24.2

%

Third-Class Passenger
Survival Rate